Bundibugyo virus, Democratic Republic of the Congo and Uganda, 2026

Latent Outbreak Visibility System (LOVS) applied to the 2026 BDBV outbreak. Ascertainment, detection depth, and pre-committed methodology calibration points, as of 9 October 2026.

Author: Frans Moore, frans@arcede.com. Source repository: https://github.com/ArcedeDev/bdbv-2026-lovs. Document is reproducible from frozen inputs via `python make_brief.py`.

Bottom line. This is a **methodology contribution** in support of the DRC Ministry of Public Health, Hygiene and Social Welfare, the Uganda Ministry of Health, the World Health Organization, and Africa CDC. **It is not a forecast, a travel advisory, or a deployment recommendation.** The 9 October 2026 snapshot indicates the public reporting picture captures only an estimated **49 to 57 percent of laboratory-confirmable cases** and that detection occurred after multiple silent transmission generations, both intrinsic to early-stage filovirus surveillance. The corridor watchlist is a pre-committed calibration test of the method's uncertainty quality, not a ranked deployment list, and on historical data the method's corridor ranking does not beat a simple proximity or caseload baseline (see calibration section).

Why a methodology brief, today. As of October 9, 2026, the most prominent public quantitative output for this outbreak is the joint WHO-Imperial College MRC GIDA report (20 May 2026 update) estimating **400-900 total cases in DRC** (values over 1,000 not excluded), via population-movement extrapolation and deaths-back-projection through the case-fatality ratio. This brief treats that estimate as a dated academic reference for outbreak size, and reproduces its deaths-back-projection (Method 2, CFR 26/33/40, at Imperial's borrowed 14-day central) to within a few cases, while re-grounding our own central doubling time from this outbreak's own confirmed-case series every cycle rather than holding it fixed: currently at most 59.9 days, estimated over a trailing incidence window and corrected for source restatements before differencing. Because confirmed cases are the product of testing effort and test positivity, and positivity is RISING (so infections are outpacing testing), this is a reported-case rate that understates true growth: read the doubling time as an upper bound and the deaths-back-projection band below as a floor. What this brief adds is the work the size estimate does not do: it is a convergence point that reconciles the scattered public sources into one provenance-tracked, source-conflict-aware view, and it publishes a reporting-completeness posterior, a pre-committed calibration set, and a cross-border corridor-risk view with date-stamped resolution. Within the archived source set for this snapshot, the other reviewed public outputs (WHO Disease Outbreak News 2026-DON602, the WHO AFRO Weekly Sitrep, Africa CDC PHECS declaration, and US CDC HAN 00530) do not include this combination. No comparable public output from the WHO Hub for Pandemic and Epidemic Intelligence in Berlin or the US CDC's Center for Forecasting and Outbreak Analytics was identified in this review as of the snapshot date. That gap is what this brief is built to fill. It complements the WHO-Imperial estimate; it does not replace it.

Authorities and standing. The Democratic Republic of the Congo (DRC) Ministry of Public Health, Hygiene and Social Welfare officially declared this outbreak on 15 May 2026 and is the lead authority on the DRC response, with the National Institute of Biomedical Research (INRB) confirming Bundibugyo virus (BDBV) by polymerase chain reaction (PCR). The Uganda Ministry of Health (MoH) is the lead authority on the Uganda response and confirmed two imported cases in Kampala on 15-16 May 2026. The World Health Organization (WHO) Director-General determined the outbreak a Public Health Emergency of International Concern (PHEIC) on 16 May 2026; WHO published the public statement on 17 May 2026. The Africa Centres for Disease Control and Prevention (Africa CDC), acting on the recommendation of the Emergency Consultative Group, declared the outbreak a Public Health Emergency of Continental Security (PHECS) on 18 May 2026. This brief is a methodology contribution in support of those authorities. It is not a substitute for them and does not speak on behalf of any of them.

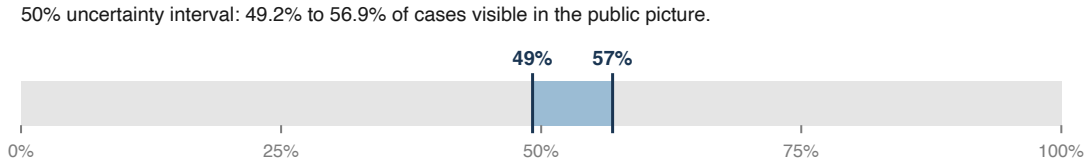
At a glance. Ebola disease caused by BDBV, a less common species of Ebola first identified in Uganda in 2007, is spreading in eastern DRC with cross-border and inter-province detection events. Public counts as of 9 October 2026: **8934 cumulative laboratory-confirmed cases** (INSP/INRB SitRep: 8914 confirmed in DRC plus 20 confirmed cases in Uganda; earlier anchors are 10 confirmed as of 17 May per WHO PHEIC after Kinshasa deconfirmation, and 30 as of 19 May per ECDC), and **4317 confirmed deaths**, including four healthcare worker deaths at Mongbwalu General Referral Hospital within a four-day span per WHO DON 602. The cumulative figures in this brief are laboratory-confirmed cases and confirmed deaths only. This brief applies the **Latent Outbreak Visibility System (LOVS)**, a stdlib-only Python pipeline that quantifies (a) the ascertainment gap, (b) the detection-depth posterior, and (c) inter-zone corridor risk under explicit calibration. The method was calibrated against the 2014 West Africa Ebola epidemic (a Zaire-species outbreak); applying it to a Bundibugyo-species outbreak introduces a species-transfer uncertainty that is reported honestly below.

What this brief is NOT. It is NOT a travel advisory. It is NOT a recommendation to restrict cross-border movement, close markets, or redirect commercial activity. It is NOT a deployment recommendation. The named corridors below are descriptive watch points for further investigation; they are NOT predictions of where the outbreak will spread. The corridor-risk numbers reflect a known model property (see methodology caveat) that overstates per-zone risk when confirmed cases are distributed across multiple health zones, as is the case here. Decisions on travel, trade, surveillance allocation, and response posture rest with the DRC Ministry of Public Health, the Uganda Ministry of Health, WHO, and Africa CDC. The pre-committed calibration points are not surveillance forecasts.

1. Ascertainment gap is wide and quantifiable

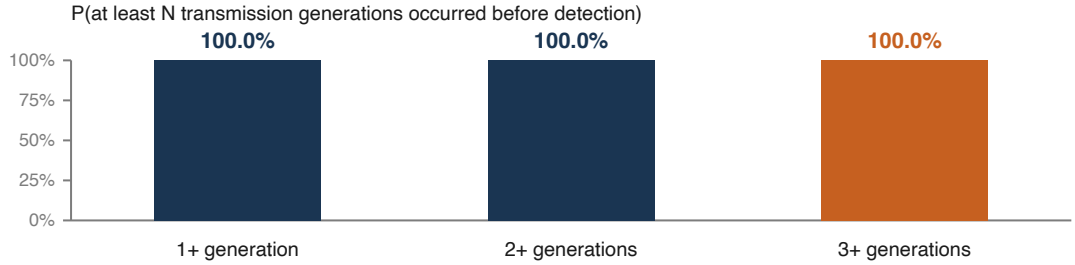
Reporting completeness (the share of underlying cases reflected in public counts), 50% uncertainty range: **[49.2%, 56.9%]**. This is consistent with what is typical of early-stage filovirus outbreaks under any surveillance system, where the gap between the visible surface and the underlying outbreak reflects the inherent reporting delay (Rosello 2015 eLife, the BDBV Isiro 2012 onset-to-notification distribution adopted as the species-matched default in the 23 May parameter audit, with Camacho 2015 PLOS Currents EBOV-Zaire retained as a faster-reporting sensitivity comparator), the historical pattern of late Bundibugyo detection (Wamala 2010 EID), and operational realities of Ituri-region surveillance: ongoing insecurity in

eastern DRC (per the Armed Conflict Location and Event Data (ACLED) project), internally displaced populations, and malaria plus other febrile, gastrointestinal, arboviral, or influenza-like illnesses that can complicate early clinical triage. The ascertainment gap is a structural feature of early outbreak reporting, not a critique of the national response. This ascertainment gap is a detection measure (its denominator is true, laboratory-confirmable infections) and should not be conflated with the source-attribution lag in the corridor section below (a spatial-allocation measure whose denominator is the cases already detected): the first asks how much of the true outbreak has been detected at all, the second asks how many detected cases have been geolocated to a specific health zone rather than held in the national unallocated residual. They are complementary layers of completeness, not competing methods.



2. Detection occurred after multiple silent transmission rounds

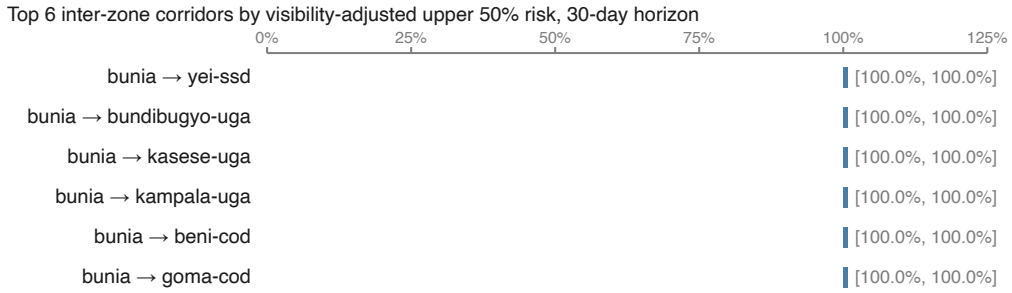
Posterior probability that at least three transmission generations (person-to-person rounds of spread) had already occurred before the outbreak became visible: **100.0%**. With 8934 laboratory-confirmed cases, the branching-process inversion places near-unity mass on three or more silent generations preceding public detection under the current R and under-ascertainment priors. Wamala 2010 and MacNeil 2010 anchor the Bundibugyo-species incubation and interval context, but the numeric bins in this chart are driven by confirmed cases, the interim R prior, and the under-ascertainment prior. The R prior remains an explicitly labeled modeling assumption because a direct Bundibugyo-virus R0 estimate has not been located.



3. Corridor watch list (descriptive, not a ranking, not a forecast)

The current 583-corridor watchlist spans 0.5-100.0% lower bounds and 1.5-100.0% upper bounds at a 30-day horizon. There is no DRC confirmed-case source-attribution lag in this cut; a future nonzero residual would be source-attribution lag, not missing cases. The watchlist separates the 8934 confirmed cases in the headline aggregate from the official per-health-zone source-load vector; the DRC national count is 8914 confirmed cases. Corridor risk now uses 8914 confirmed cases that are officially zone-attributed across 65 official source zones, rather than applying the headline aggregate to every source zone. The DRC residual is 0 confirmed cases: every DRC confirmed case is officially zone-attributed, so none is unallocated. Uganda's 20 confirmed cases are country-scoped and never assigned to DRC zones. Per-zone confirmed deaths trail case attribution by roughly 1-3 weeks while the INRB clinical review queue closes, so the per-zone confirmed-deaths figure is a lower bound. The remaining clustering is still a limitation signal: no single corridor stands clearly above the others; on historical data the method does not out-rank a simple proximity or caseload baseline (see calibration section).

Methodology caveat (load-bearing). The snapshot carries two different count concepts. The headline public count is 8934 confirmed cases as of 9 October 2026 (INSP/INRB SitRep: 8914 confirmed in DRC plus 20 confirmed cases in Uganda). The corridor source-load vector is spatially attributed: the INRB-UMIE/INSP per-health-zone series attributes 8914 confirmed DRC cases across 65 official source zones as of 9 October. The corridor model uses that per-zone vector because it is the newest officially zone-attributed table available in the archive. It does not scale the vector up to the 8934 country-scope headline aggregate: that aggregate includes Uganda's 20 country-scoped confirmed cases, which belong to no DRC zone, and the DRC residual against the 8914 DRC national count is 0 confirmed cases. Separately, the corridor model's gravity parameters (the population, road, healthcare-distance, and conflict exponents and the clamp) are transparent engineering heuristics, not fitted to a mobility dataset: Backer & Wallinga 2016 is the West Africa 2014 validation substrate and supports the broad gravity-type model family, but it does not source-fit the current LOVS constants; the public audit trail records that limitation without exposing internal evidence-chain identifiers.



Read as a watch list, not a ranking decisive enough to direct deployment.

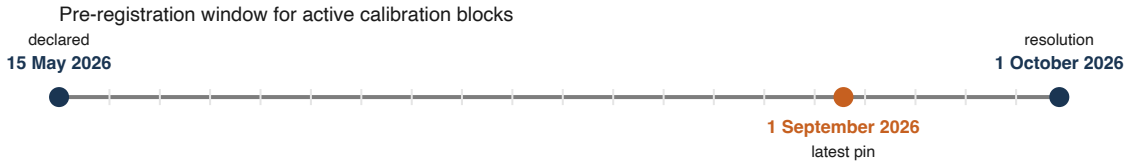
4. Calibration block carried by the 9 October 2026 snapshot

This 9 October 2026 snapshot carries forward the 20 May 2026 calibration block unchanged; it does not pin a new calibration block. Future snapshots can append new blocks with their own pin date, horizon, and resolution timestamp without moving this block. The named corridors below were pre-committed as calibration points for the underlying method. Calibration is narrower than the descriptive watchlist: a corridor only becomes a calibration point when it is pinned before outcome assessment with a fixed source, target, uncertainty range, horizon, and resolution timestamp. The remaining watchlist corridors stay visible as descriptive signals; they are not counted as extra independent calibration evidence unless a later pre-publication block explicitly pins them. Each pinned point is paired with the model's ascertainment-adjusted 50% uncertainty range for a binary outcome at a 30-day horizon from its pin date. As of this 9 October 2026 snapshot, 142 days have elapsed since the pin and 0 days remain until resolution. The clock equation is `remaining_days = date(resolves_at) − date(as_of)`. Outcomes will be scored against publicly available reports from the DRC Ministry of Public Health, the Uganda Ministry of Health, the WHO, and Africa CDC, in order to evaluate whether the model's uncertainty ranges contained the observed outcome and to update the historical calibration. **These calibration points are NOT recommendations for the active public-health response.**

What each calibration point claims, in plain language:

Calibration point	Block / design role	Plain-language statement
Calibration point 1	2026-05-20	At least one new laboratory-confirmed BDBV case appears in Kampala (Uganda) between 20 May 2026 and 19 June 2026, given continued reporting from Bunia Health Zone (Ituri Province, DRC). Model's ascertainment-adjusted 50% uncertainty range: **[22.9%, 52.3%]**.
Calibration point 2	2026-05-20	At least one new laboratory-confirmed BDBV case appears in Bundibugyo District (Uganda) between 20 May 2026 and 19 June 2026, given continued reporting from Rwampara Health Zone (Ituri Province, DRC). Model's ascertainment-adjusted 50% uncertainty range: **[22.7%, 52.3%]**.
Calibration point 3	2026-05-20	At least one new laboratory-confirmed BDBV case appears in Beni Health Zone (North Kivu Province, DRC) between 20 May 2026 and 19 June 2026, given continued reporting from Mongbwalu Health Zone (Ituri Province, DRC). Model's ascertainment-adjusted 50% uncertainty range: **[21.8%, 52.2%]**.
Calibration point 4	2026-05-20	At least one new laboratory-confirmed BDBV case appears in Kasese District (Uganda) between 20 May 2026 and 19 June 2026, given continued reporting from Rwampara Health Zone (Ituri Province, DRC). Model's ascertainment-adjusted 50% uncertainty range: **[20.9%, 51.5%]**.
Calibration point 5	2026-05-21 relative high / cross-border / watchlist high	At least one new laboratory-confirmed BDBV case appears in Kasese District (Uganda) between 21 May 2026 and 20 June 2026, given continued reporting from Bunia Health Zone (Ituri Province, DRC). Model's ascertainment-adjusted 50% uncertainty range: **[22.0%, 55.3%]**.
Calibration point 6	2026-05-21 relative high / in-country / likely positive	At least one new laboratory-confirmed BDBV case appears in Beni Health Zone (North Kivu Province, DRC) between 21 May 2026 and 20 June 2026, given continued reporting from Bunia Health Zone (Ituri Province, DRC). Model's ascertainment-adjusted 50% uncertainty range: **[24.3%, 52.5%]**.
Calibration point 7	2026-05-21 relative mid / cross-border / likely positive	At least one new laboratory-confirmed BDBV case appears in Kampala (Uganda) between 21 May 2026 and 20 June 2026, given continued reporting from Rwampara Health Zone (Ituri Province, DRC). Model's ascertainment-adjusted 50% uncertainty range: **[21.8%, 51.5%]**.
Calibration point 8	2026-05-21 relative mid / cross-border / watchlist mid	At least one new laboratory-confirmed BDBV case appears in Kasese District (Uganda) between 21 May 2026 and 20 June 2026, given continued reporting from Mongbwalu Health Zone (Ituri Province, DRC). Model's ascertainment-adjusted 50% uncertainty range: **[23.6%, 51.9%]**.
Calibration point 9	2026-05-21 relative mid / in-country / likely positive	At least one new laboratory-confirmed BDBV case appears in Beni Health Zone (North Kivu Province, DRC) between 21 May 2026 and 20 June 2026, given continued reporting from Rwampara Health Zone (Ituri Province, DRC). Model's ascertainment-adjusted 50% uncertainty range: **[23.6%, 51.8%]**.
Calibration point 10	2026-05-21 relative mid / cross-border / blindspot watch	At least one new laboratory-confirmed BDBV case appears in Arua District (Uganda) between 21 May 2026 and 20 June 2026, given continued reporting from Bunia Health Zone (Ituri Province, DRC). Model's ascertainment-adjusted 50% uncertainty range: **[23.6%, 52.0%]**.

Calibration point	Block / design role	Plain-language statement
Calibration point 11	2026-05-21 relative low / cross-border / likely negative	At least one new laboratory-confirmed BDBV case appears in Arua District (Uganda) between 21 May 2026 and 20 June 2026, given continued reporting from Rwampara Health Zone (Ituri Province, DRC). Model's ascertainment-adjusted 50% uncertainty range: **[22.1%, 49.4%]**.
Calibration point 12	2026-05-21 relative low / cross-border / likely negative	At least one new laboratory-confirmed BDBV case appears in Nebbi District (Uganda) between 21 May 2026 and 20 June 2026, given continued reporting from Mongbwalu Health Zone (Ituri Province, DRC). Model's ascertainment-adjusted 50% uncertainty range: **[22.2%, 51.0%]**.
Calibration point 13	2026-05-26 relative mid / in-country / watchlist mid	At least one new laboratory-confirmed BDBV case appears in goma-cod between 26 May 2026 and 25 June 2026, given continued reporting from Bunia Health Zone (Ituri Province, DRC). Model's ascertainment-adjusted 50% uncertainty range: **[10.2%, 26.7%]**.
Calibration point 14	2026-05-26 relative high / in-country / watchlist high	At least one new laboratory-confirmed BDBV case appears in goma-cod between 26 May 2026 and 25 June 2026, given continued reporting from Rwampara Health Zone (Ituri Province, DRC). Model's ascertainment-adjusted 50% uncertainty range: **[20.3%, 45.9%]**.
Calibration point 15	2026-05-26 relative mid / in-country / watchlist mid	At least one new laboratory-confirmed BDBV case appears in goma-cod between 26 May 2026 and 25 June 2026, given continued reporting from Mongbwalu Health Zone (Ituri Province, DRC). Model's ascertainment-adjusted 50% uncertainty range: **[9.6%, 25.2%]**.
Calibration point 16	2026-06-04 relative high / cross-border / watchlist mid	At least one new laboratory-confirmed BDBV case appears in yei-ssd between 4 June 2026 and 4 July 2026, given continued reporting from Bunia Health Zone (Ituri Province, DRC). Model's ascertainment-adjusted 50% uncertainty range: **[38.9%, 73.4%]**.
Calibration point 17	2026-06-04 relative low / cross-border / blindspot watch	At least one new laboratory-confirmed BDBV case appears in yei-ssd between 4 June 2026 and 4 July 2026, given continued reporting from aru. Model's ascertainment-adjusted 50% uncertainty range: **[1.2%, 3.2%]**.
Calibration point 18	2026-06-04 relative high / in-country / watchlist mid	At least one new laboratory-confirmed BDBV case appears in kisangani-cod between 4 June 2026 and 4 July 2026, given continued reporting from Bunia Health Zone (Ituri Province, DRC). Model's ascertainment-adjusted 50% uncertainty range: **[36.6%, 71.3%]**.
Calibration point 19	2026-06-04 relative mid / in-country / watchlist mid	At least one new laboratory-confirmed BDBV case appears in kisangani-cod between 4 June 2026 and 4 July 2026, given continued reporting from Mongbwalu Health Zone (Ituri Province, DRC). Model's ascertainment-adjusted 50% uncertainty range: **[23.3%, 52.3%]**.
Calibration point 20	2026-09-01 saturated / cross-border / model saturation test	At least one new laboratory-confirmed BDBV case appears in Arua District (Uganda) between 1 September 2026 and 1 October 2026, given continued reporting from Bunia Health Zone (Ituri Province, DRC). Model's ascertainment-adjusted 50% uncertainty range: **[100.0%, 100.0%]**.
Calibration point 21	2026-09-01 saturated / cross-border / model saturation test	At least one new laboratory-confirmed BDBV case appears in Bundibugyo District (Uganda) between 1 September 2026 and 1 October 2026, given continued reporting from Bunia Health Zone (Ituri Province, DRC). Model's ascertainment-adjusted 50% uncertainty range: **[100.0%, 100.0%]**.
Calibration point 22	2026-09-01 saturated / cross-border / model saturation test	At least one new laboratory-confirmed BDBV case appears in Kampala (Uganda) between 1 September 2026 and 1 October 2026, given continued reporting from Bunia Health Zone (Ituri Province, DRC). Model's ascertainment-adjusted 50% uncertainty range: **[100.0%, 100.0%]**.
Calibration point 23	2026-09-01 saturated / cross-border / model saturation test	At least one new laboratory-confirmed BDBV case appears in Kasese District (Uganda) between 1 September 2026 and 1 October 2026, given continued reporting from Bunia Health Zone (Ituri Province, DRC). Model's ascertainment-adjusted 50% uncertainty range: **[100.0%, 100.0%]**.
Calibration point 24	2026-09-01 saturated / cross-border / model saturation test	At least one new laboratory-confirmed BDBV case appears in Nebbi District (Uganda) between 1 September 2026 and 1 October 2026, given continued reporting from Bunia Health Zone (Ituri Province, DRC). Model's ascertainment-adjusted 50% uncertainty range: **[100.0%, 100.0%]**.
Calibration point 25	2026-09-01 saturated / cross-border / model saturation test	At least one new laboratory-confirmed BDBV case appears in yei-ssd between 1 September 2026 and 1 October 2026, given continued reporting from Bunia Health Zone (Ituri Province, DRC). Model's ascertainment-adjusted 50% uncertainty range: **[99.9%, 100.0%]**.



Forecasts scored at resolution against public WHO, Africa CDC, DRC MoH, Uganda MoH announcements.

Historical calibration on the 2014 West Africa Ebola epidemic

The method has been tested retrospectively against the 2014 West Africa Ebola epidemic (Backer & Wallinga 2016 substrate, 62 prefectures across 74 weeks), where the eventual outcomes are public knowledge. Three runs are reported: without local context, with country-level local context (population density, road density, healthcare access, conflict intensity), and with district-level local context. Adding local context at country level tightens uncertainty (interval score down by roughly 35%); it does NOT improve discrimination of individual corridors above chance. The 2014 substrate was a Zaire-species outbreak; the method's calibration on Zaire data transfers to a Bundibugyo-species outbreak with a species-transfer uncertainty that is not separately quantified here.

Metric	No local context	Country-level	District-level	Change vs country
Brier score (probability accuracy; lower is better)	0.0586	0.0590	0.0590	+0.0000
Interval score (Bracher 2021; uncertainty quality; lower is better)	0.1002	0.0649	0.0649	+0.0000
Calibration error (predicted vs observed; lower is better)	0.0391	0.0500	0.0500	+0.0000

Finding. Country-level local context tightens uncertainty by approximately 35% but does not lift discrimination above chance, and district-level granularity does not add further. Candidate next-levers, ranked by expected impact: mobility data (Wesolowski 2016 Journal of Infectious Diseases), time-varying context (real-time political-violence intensity from ACLED and real-time MoH reporting cadence), re-engineering how the four context variables combine, and a richer transmission model. The calibration-error increase from 0.039 to 0.050 between no-context and context runs is small and most likely an artifact of the small evaluation set (five as-of dates).

Robustness check (skill, discrimination, calibration). A rolling-origin backtest across a pre-registered grid of as-of weeks, reproducible via `robustness_backtest.py`, refines the table above. At the early checkpoints the method ranks which prefectures appear next with a ROC AUC of 0.72, above the 0.50 chance level, but a distance-only baseline (0.73) and a source-load-only baseline (0.72) rank just as well: the gravity and covariate machinery add no ranking value, so the discrimination that exists is the spatial autocorrelation of the epidemic rather than the model. Against a base-rate (climatology) reference the Brier skill score is -0.03 (95 percent interval -0.48 to 0.09); across the full grid no configuration shows a skill interval above zero at any window, so the method has no positive calibration skill beyond predicting the base rate. Intervals are from a target-prefecture clustered bootstrap. These are 2014 West Africa (Zaire-species) results and are not a portable skill claim for a Bundibugyo-species outbreak.

If you have point-of-care data

This brief reports method estimates from publicly aggregated reporting only. If you are working directly in the affected zones, you almost certainly hold information that is privileged, time-sensitive, and not appropriate for a public repository. **Please do not paste line-list rows, GPS-tagged case locations, sequencing reads, or any identifying detail into a public GitHub issue.** You can reach me directly at frans@arcede.com if any of the following would help your work.

- **Onset-date line-list extract (de-identified).** Even a partial onset-date histogram for one health zone substantially narrows the latent-active-chains plausibility interval emitted by Module D.
- **Zone-attributed case counts.** The load-bearing simplification flagged in Panel 3 is that the full confirmed count is contributed to each named source zone. A mapping of `{zone_id: count}` for the affected districts is the largest single discrimination lever the method is missing.
- **Validated zone GPS centroids.** The published map ships verified centroids for zones currently in scope (`data/zones.json`). For zones we may have missed, a centroid plus a one-sentence rationale is enough to extend the corridor model.
- **Mobility traces or transport-flow snapshots.** Call-detail-record summaries or surveyed transport flows, even at admin-2 aggregation, are the documented next-lever for moving the method above chance discrimination (Wesolowski 2015 PNAS).
- **Case-confirmation latency.** Field-observed delays (sample collection to PCR result, in days) are a direct prior update for Module C, replacing the historical onset-to-notification prior (Rosello 2015 BDBV Isiro 2012) currently used as the default.

Contributions that land in the repository are cited and timestamped. If you prefer to remain unnamed, I can credit you by initials or pseudonym at the time of contribution.

Known blindspots and calibration design notes

- **Mahagi as a source zone is still outside the model, while Arua/Nebbi are now represented as target-side calibration/watch corridors.** Mahagi (DRC) and Goli (Uganda) form one of DRC's busiest land border crossings on the East African Northern Corridor, with 95,000+ refugees at the Rhino Camp settlement near Arua (UNHCR, late-2025). The

21 May 2026 revision added Arua District and Nebbi District to the candidate-target watch set, and the unpublished 21 May calibration block pins selected Arua/Nebbi target corridors before publication. The 22 May WHO DG and IHR sources were rechecked for this blindspot and still do not name Mahagi health zone as case-affected. Mahagi (DRC) is therefore still not modeled as a source zone; `snapshot_sensitivity.py` shows where the omitted Mahagi-to-Arua corridor would rank under an explicit equal-burden counterfactual.

- **The active calibration corridors preserve their original pinned probability band (1.2 to 100.0% lower, 3.2 to 100.0% upper).** The May 20 and May 21 calibration blocks are not re-derived after the May 22 spatial correction; that is intentional. The current watchlist now uses zone-attributed source loads, while the calibration ledger remains an immutable record of what was pre-committed before the correction.
- **The corridor mongbwalu-to-beni-cod may be confounded.** Beni Health Zone sits in North Kivu Province, which the 18 May Africa CDC reporting identifies as already part of the outbreak footprint. A positive resolution on this corridor may simply reflect Beni being part of the active outbreak rather than source-zone-to-target-zone transmission.
- **Conflict-state coverage is qualitative.** The brief invokes CODECO and ADF activity in Ituri/North Kivu and 7.3 million IDPs in eastern DRC as descriptive context, but the per-zone conflict-intensity input is the 2014 West Africa ACLED snapshot used in the Mode A historical calibration. A 2026 ACLED snapshot for eastern DRC is a documented next-lever.

What this brief does NOT claim

- **Not a forecast.** The pre-committed calibration points exist to evaluate the method's uncertainty quality at resolution. They are not surveillance recommendations and not deployment recommendations.
- **Not a critique of the national response.** Ascertainment gaps and late detection of filoviruses are intrinsic to the pathogen and to the operational context (security, displacement, co-circulating pathogens), not to the speed of any specific national response. The DRC Ministry of Public Health declared on 15 May, the Uganda Ministry of Health confirmed imported cases on 15-16 May, INRB confirmed BDBV by PCR within days. This brief takes the national declarations as the authoritative timeline.
- **Does not replace field epidemiology.** Line-listing, contact tracing, genomic sequencing, and clinical reasoning are where outbreak control happens.
- **Species-transfer uncertainty is not separately quantified.** The historical calibration substrate is a Zaire-species outbreak; transfer to Bundibugyo carries unquantified uncertainty in the priors and the corridor model.
- **Numbers are prior-dominated at this case count.** The ascertainment and transmission-depth posteriors are heavily informed by the prior delay and transmission distributions, not by data alone. In particular, the branching-process reproduction prior is an interim modeling assumption: no Bundibugyo-virus-specific basic reproduction number has been estimated in the published literature (Van Kerkhove 2015), so the detection-depth result is prior-driven and species-transferred, not a data-driven BDBV estimate. Sensitivity analyses across alternative priors are a recommended next step.
- **Open to independent replication.** Code is Apache 2.0 licensed; original authored methodology, prose, schema, and derived artifacts are CC BY 4.0; third-party source material and extracted publisher-owned tables retain their original terms. The method is described in `CITATIONS.md`. Independent replication is welcomed.

Citation. Moore F. *Bundibugyo virus, DRC and Uganda, 2026: surveillance methodology brief.* Released 2026-05. https://github.com/ArcedeDev/bdbv-2026-lovs. | **Live data:** WHO Disease Outbreak News item 2026-DON602, WHO African Region Weekly External Situation Report 01 (data as of 18 May 2026), Africa CDC PHECS declaration (18 May 2026), and aggregated public reporting through 9 October 2026. | **Methodology citations:** Wamala 2010 Emerging Infectious Diseases (EID), MacNeil 2010 EID, Albariño 2013 Virology, Faye 2015 Lancet ID, Backer & Wallinga 2016 PLOS Computational Biology, Bracher 2021 PLOS Computational Biology (full list at `CITATIONS.md`). | **License.** Code Apache 2.0; original authored methodology/prose/schema/derived artifacts CC BY 4.0; third-party source material keeps publisher terms; WHO content Creative Commons BY-NC-SA 3.0 IGO. | **Pre-commitment timestamp:** verifiable via the git commit history of `data/calibration_ledger.json` in the source repository (first pinned on 2026-05-20, before the 2026-06-19 resolution window). | **Resolution:** 2026-06-19T23:59:59Z.